

Zhankun Luo

Machine Learning Theory & Optimization

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RESEARCH PROFILE

I develop and analyze algorithms for stochastic optimization and statistical learning, with an emphasis on convergence and finite-sample guarantees. My work spans variance reduction and randomized methods, distributed optimization under stochastic constraints, and the dynamics and statistical guarantees of latent-variable learning. My ongoing research investigates oracle-complexity lower and upper bounds for nonconvex optimization under state-dependent heavy-tailed gradient noise.

EDUCATION

Purdue University

Ph.D. in Electrical and Computer Engineering

Advisor: Prof. Abolfazl Hashemi; Machine Intelligence & Networked Data Science (MINDS) Group.

Present

West Lafayette, IN

Purdue University Northwest

M.S. in Electrical and Computer Engineering

2021

Hammond, IN

Beijing Institute of Technology

B.S. in Telecommunication Engineering

2019

Beijing, China

SELECTED THEORETICAL CONTRIBUTIONS

Variance reduction and randomized algorithms

- › Developed a unified high-probability analysis of stochastic variance-reduced estimators using a new Freedman-type inequality, including mirror-descent guarantees in Banach spaces and improved oracle complexity for expectation-constrained optimization. [paper](#) [poster](#) [7] [NeurIPS 2026]
- › Developed unbiased randomized midpoint estimators that correct discretization bias in nonlinear extragradient updates; used antithetic sampling to cancel first-order variance terms and established guarantees for root finding, constrained variational inequalities, and smooth convex-concave games. [paper](#) [5]
- › Formulated strong antithetic maps and indices to maximize negative dependence while preserving a target law, connecting optimal antithetic coupling to geometric optimal transport for Monte Carlo integration, approximation, and stochastic optimization. [slides](#) [2]

Distributed stochastic minimax optimization with constraints

- › Designed a single-loop, primal-only Softmax-weighted switching-gradient method for worst-client optimization under stochastic constraints; established convergence under full and partial client participation with logarithmic high-probability dependence. [paper](#) [poster](#) [code](#) [8] [UAI 2026]

Dynamics and statistical guarantees for latent-variable models

- › Derived explicit population EM updates across all signal-to-noise regimes; proved that noiseless iterates lie on a cycloid, estimated the super-linear convergence exponent, and improved finite-sample statistical error bounds. [paper](#) [poster](#) [11] [ICML 2024]
- › Extended the trajectory analysis to unknown mixing weights and regression parameters, establishing linear-to-quadratic population convergence and arbitrary-initialization finite-sample guarantees. [paper](#) [code](#) [9] [IEEE TIT R&R; revised manuscript submitted]
- › Characterized overspecified EM with jointly estimated regression parameters and mixing weights, proving distinct linear and sublinear regimes and deriving iteration, sample, and statistical-accuracy guarantees in population and finite samples. [paper](#) [code](#) [10] [TMLR 2026]
- › Proved consistency and asymptotic normality with maximum-likelihood covariance for path-integrated score matching in mixed linear regression with unknown mixing weights under suitable diffusion schedules; derived exact fixed-noise decompositions linking the score-matching loss and gradients to EM operators. [paper](#) [6]

Stochastic optimization under state-dependent heavy-tailed noise

- › Investigating oracle-complexity lower bounds and algorithmic upper bounds for nonconvex optimization under state-dependent bounds on gradient-noise moments. [slides](#) [1]

PUBLICATIONS & MANUSCRIPTS

- [1] **Z. Luo**, A. Fazla, A. Upadhyay, S. B. Moon, B. Shin, and A. Hashemi. “Stochastic Nonconvex Optimization with Heavy-Tailed Noise: Distance and Objective Variation.” *Manuscript in preparation*.
- [2] A. Hashemi, **Z. Luo**, and A. Makur. “What Constitutes a Multivariate Antithetic Pair? Representation and Learning via Optimal Transport.” *Submitted to AISTATS. Authors listed alphabetically by surname*.
- [3] R. Wu, **Z. Luo**, A. Hashemi, and A. Wang. “A Communication-Efficient Distributed Augmented Lagrangian Method for Nonsmooth Regularized Learning.” *Submitted to AISTATS*.
- [4] A. Upadhyay, S. B. Moon, **Z. Luo**, and A. Hashemi. “Variance Reduction for Stochastic Switching in Functionally Constrained Federated Optimization.” *Submitted to AISTATS*.
- [5] **Z. Luo**, M. B. Sahin, A. Upadhyay, B. Sharif, and A. Hashemi. “RAMPAGE: RAndomized Mid-Point for debiAsed Gradient Extrapolation.” *Submitted to AISTATS*. [paper](#)
- [6] **Z. Luo** and A. Hashemi. “Connecting Score Matching, Maximum Likelihood, and Expectation-Maximization in Mixed Linear Regression.” *Submitted to TMLR*. [paper](#)
- [7] **Z. Luo**, A. Upadhyay, M. B. Sahin, S. B. Moon, A. Makur, and A. Hashemi. “Unified High-Probability Analysis of Stochastic Variance-Reduced Estimation.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. [paper](#) [poster](#)
- [8] **Z. Luo**, A. Upadhyay, S. B. Moon, and A. Hashemi. “First-Order Softmax Weighted Switching Gradient Method for Distributed Stochastic Minimax Optimization with Stochastic Constraints.” *Uncertainty in Artificial Intelligence (UAI)*, 2026. [paper](#) [poster](#) [code](#)
- [9] **Z. Luo** and A. Hashemi. “Structural Properties, Cycloid Trajectories and Non-Asymptotic Guarantees of EM Algorithm for Mixed Linear Regression.” *Revised manuscript submitted to IEEE TIT following a revise-and-resubmit decision*. [paper](#) [code](#)
- [10] **Z. Luo** and A. Hashemi. “Characterizing Evolution in Expectation-Maximization Estimates for Overspecified Mixed Linear Regression.” *Transactions on Machine Learning Research (TMLR)*, 2026. [paper](#) [code](#)
- [11] **Z. Luo** and A. Hashemi. “Unveiling the Cycloid Trajectory of EM Iterations in Mixed Linear Regression.” *International Conference on Machine Learning (ICML)*, 2024. [paper](#) [code](#)
- [12] E. Cai, **Z. Luo**, S. Baireddy, J. Guo, C. Yang, and E. J. Delp. “High-Resolution UAV Image Generation for Sorghum Panicle Detection.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2022. [paper](#) [video](#)
- [13] N. J. Walla, **Z. Luo**, B. Chen, Y. Krotov, and C. Q. Zhou. “Smart Ladle: AI-Based Tool for Optimizing Caster Temperature.” *Iron & Steel Technology Conference (AISTech)*, 2021. [paper](#) [poster](#)
- [14] **Z. Luo**. “Structured Light Vision Systems Using a Robust Laser Stripe Segmentation Method.” M.S. thesis, Purdue University, 2021. [paper](#)
- [15] **Z. Luo**, Y. Zhang, and L. Tan. “Multi-Level Random Sample Consensus Method for Improving Structured Light Vision Systems.” *IEEE UEMCON*, 2020. [paper](#)
- [16] Y. Zhang, **Z. Luo**, J. Hou, L. Tan, and X. Guo. “Computer Vision Techniques for Improving Structured Light Vision Systems.” *IEEE EIT*, 2020. [paper](#)

RESEARCH EXPERIENCE

Research Assistant, MINDS Group, Purdue University

PI: [Prof. Abolfazl Hashemi](#)

Aug. 2025 – Present

West Lafayette, IN

- › Design variance-reduced and randomized methods for stochastic minimax and expectation-constrained optimization, with convergence and oracle-complexity guarantees.
- › Develop federated-optimization algorithms for full and partial client participation under communication and resource constraints within the NSF project [Learning through the Air: Cross-Layer UAV Orchestration for Online Federated Optimization](#).

Research Assistant, VIPER Laboratory, Purdue University

PIs: [Prof. Edward J. Delp](#) and [Prof. Fengqing M. Zhu](#)

Aug. 2021 – May 2022

West Lafayette, IN

- › Investigated hierarchy-aware learning for fine-grained food recognition in the [Technology Assisted Dietary Assessment \(TADA\)](#) project and generated labeled UAV imagery with image-to-image GANs for the [PhenoSorg plant-phenotyping](#) project.

Research Assistant, CIVS, Purdue University Northwest

PIs: [Prof. Chenn Zhou](#), [Prof. Bin Chen](#), and [Prof. Lizhe Tan](#)

Feb. 2020 – May 2021

Hammond, IN

- › Built DNN and LightGBM models for the [Smart Ladle steel casting-temperature predictor](#) using process data from SQL databases; developed robust multi-camera, multi-laser algorithms for structured-light metrology.

TEACHING EXPERIENCE

Teaching Assistant, Purdue University ECE

ECE 69500: *Optimization for Deep Learning*; ECE 20001/20002: *EE Fundamentals*

May 2022 – Aug. 2025

West Lafayette, IN

- › **ECE 69500**: Supported instruction in stochastic gradient methods, generalization, and distributed learning; held office hours and graded homework and exams. [slides](#) [video](#)
- › **ECE 20001/20002**: Led office hours and help sessions, graded assessments, coordinated TA activities, and compiled an ECE 20001 practice-problem collection. [problems](#) [video](#)

SELECTED PRESENTATIONS

- › **First-Order Softmax Weighted Switching Gradient Method for Distributed Stochastic Minimax Optimization with Stochastic Constraints.** Online presentation, UAI 2026, Aug. 2026. [poster](#)
- › **Poster presentation**, Midwest Machine Learning Symposium, Purdue University, Jun. 2026. [event](#)
- › **Strong Antithetic Variates: Theory and Applications.** MINDS Group presentation, Purdue University, Nov. 2025. [slides](#)
- › **High-Resolution UAV Image Generation for Sorghum Panicle Detection.** Oral presentation, Agriculture-Vision Workshop at CVPR 2022, New Orleans, LA, Jun. 19, 2022. [video](#)

PROFESSIONAL SERVICE

Journal reviewer: IEEE Transactions on Signal Processing; Stochastic Systems.

Conference reviewer: NeurIPS; ICLR; UAI; AISTATS; AAAI; IEEE ISIT; IEEE ICME; CVIS.

Memberships: IEEE Young Professionals; IEEE Signal Processing Society; IEEE Computer Society.

SELECTED RESEARCH SOFTWARE

SoftmaxSGM	Implementation of the UAI 2026 Softmax-weighted switching-gradient method.
em_overspecified_mlr	Numerical experiments for the TMLR 2026 analysis of overspecified EM.
cycloid_em_mlr	Numerical experiments and visualizations for the ICML 2024 cycloid analysis.
smart_ladle	LightGBM casting-temperature predictor with SQL integration, tested and deployed at Steel Dynamics' Butler Division. SDI Butler
hierarchy_classification	PyTorch reimplement of hierarchy-aware food classification and embedding.
Kmeans-tsne-visualization	C++/Qt/OpenGL application for interactive visualization of t-SNE embeddings and k-means iterations.

LEADERSHIP, MENTORING & OUTREACH

- › **Graduate Student Mentor, English Training in Engineering (ETIE):** designed machine-learning and computer-vision project frameworks and mentored undergraduate research in sequential prediction and 3D reconstruction.
- › **Coordinator and Host, Science Communication Contest:** coordinated participants and judges and organized a science communication program serving more than 3,600 students.
- › **Team Leader, Mathematical Modeling Contests:** led model development and implementation for national and international competitions, earning a CUMCM Second Prize and an ICM Honorable Mention.

SELECTED AWARDS & HONORS

NeurIPS Conference Travel Grant	2025
AIST Hunt-Kelly Outstanding Paper Award, third place	2022
AIST Digitalization Applications Technology Best Paper Award	2021
Innovative Practice Outstanding Student (top 5%)	2019
National Third Prize, Energy Saving and Emission Reduction Competition	2017
Interdisciplinary Contest in Modeling, Honorable Mention	2017
Contemporary Undergraduate Mathematical Contest in Modeling, Second Prize	2016

TECHNICAL BACKGROUND

Theory: stochastic, constrained, minimax, and nonconvex optimization; variance reduction; concentration and martingale inequalities; oracle complexity bounds; non-asymptotic statistics; federated learning; dynamical systems; optimal transport.

Programming & scientific computing: Python, Mathematica, MATLAB, C/C++, SQL, Bash, C#, Java; NumPy, SciPy, pandas.

Machine learning & vision: PyTorch, TensorFlow, LightGBM, scikit-learn, OpenCV; multi-view geometry, camera calibration.

Research software & tools: Linux, Git, Docker, CMake, gdb, Qt, OpenGL, Unity, $\LaTeX$.

SELECTED COURSEWORK

Real Analysis; Linear Algebra with Applications; Introduction to Applied Mathematics; Introduction to Mathematical Statistics
Optimization for Deep Learning; Statistical Machine Learning; Probabilistic Machine Learning; Reinforcement Learning
Random Variables and Signals; Estimation Theory; Pattern Recognition and Decision Making
Model-Based Image and Signal Processing; Digital Signal Processing I; Digital Image Processing I
Digital Control Systems Analysis and Design; Object-Oriented Design, Analysis and Programming